ALY 3510 AI Foundations: An Interdisciplinary Approach

Week 7:
Feedback (week 6)

Session/Section: Fall 2026
Instructor: Vladimir Shapiro



Northeastern University

Feedback - Week 6

Reflection Journal



Going well

- Good understanding of the main concepts
- Good research by some
- Pengyu: "What surprised me most is zero-shot translation: a model can translate between two languages it was never directly trained on, because both connect to the same meaning space."
- Other clues (previous students)
  - Speech tone, facial expression out of the MLP scope per se.
- Previous student: "A phrase that sounds obviously sarcastic to one group may be interpreted literally by another… sarcasm is not only a technical problem for AI, but also a human challenge "
- Yessenia: "Even people can misunderstand sarcasm sometimes, so I can imagine how difficult it can be for AI."
- Don't underestimate AI, see below/

Feedback - Week 6

Reflection Journal



Going well

- John: "Thanks for getting that delivery here" from a customer - You wouldn't really know if it was definitely sarcasm or genuine unless you knew the person well."
- Lam Ky "I switch between Vietnamese and English every day, so I never thought much about how difficult this could be for AI."
- Brandon: "Even humans sometimes misunderstand it, so expecting machines to get it right every time is unrealistic."
- Good visual (Nathaniel, Jennifer)
- Jorgo: "Meaning becomes geometry"



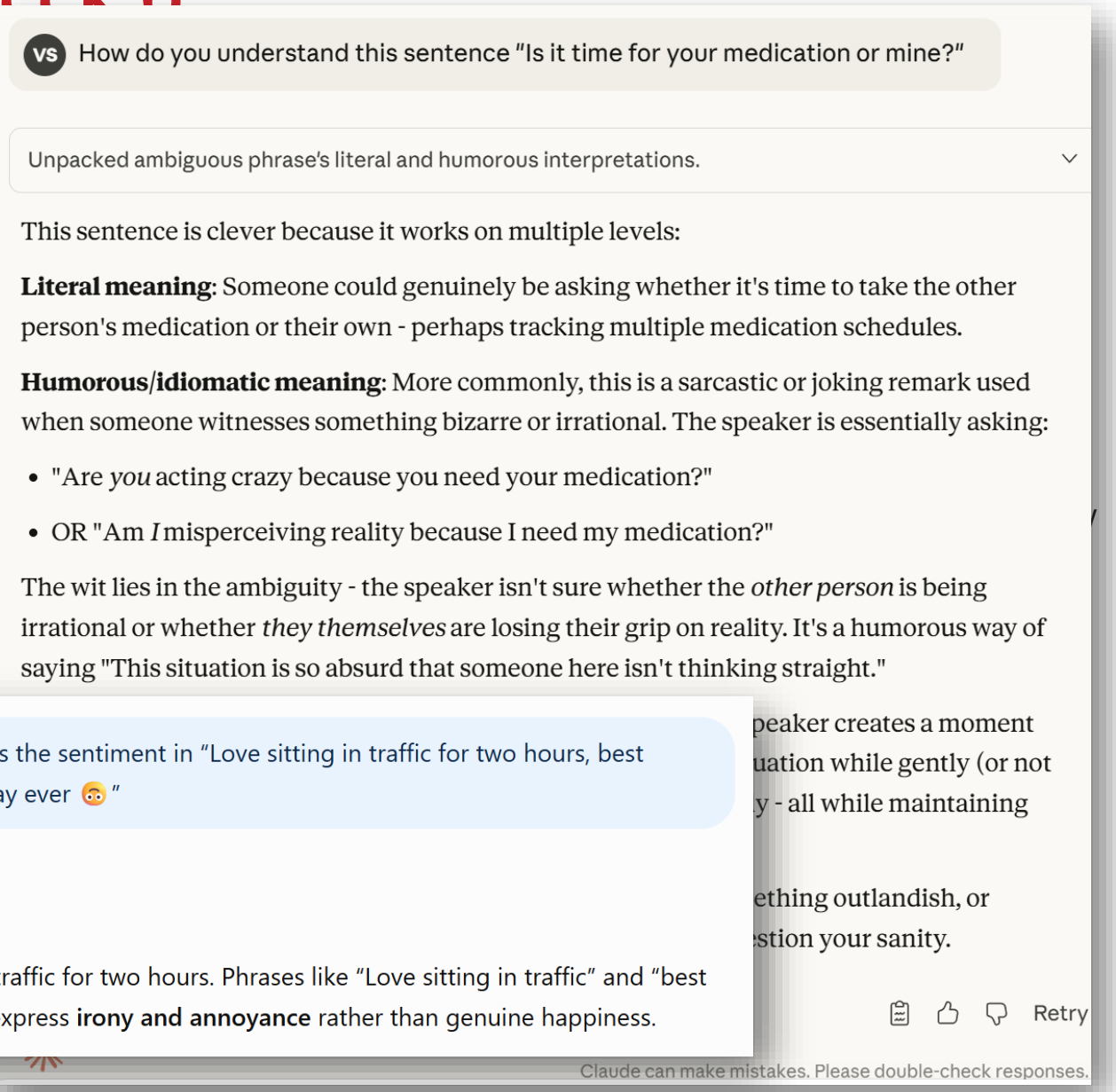
Figure 1. Note. Example sentence from AWS (n.d.); concepts adapted from this weeks Lecture 6, slides 10 and 27. Diagram generated with Claude (Opus 5.5 version; Anthropic, 2026) and reviewed by the author.

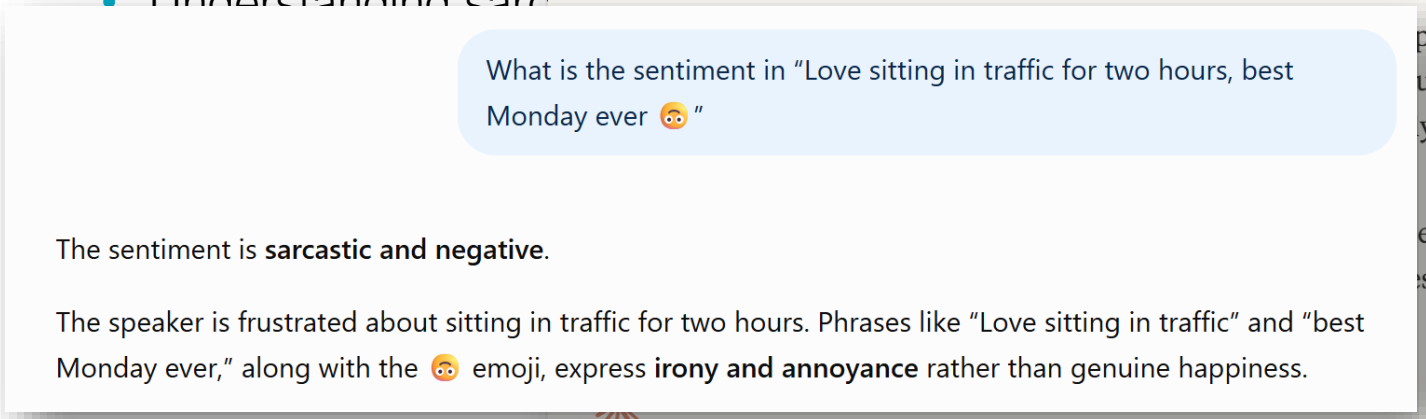
Feedback - Week 6

Reflection Journal

To improve

- It would be beneficia
- Using sources from
- “NLP will never be
  about humans?
- Understanding sarc





Feedback - Week 6

Assignment



Going well:

- Labor-intensive, intellectually challenging assignment… High grades, well deserved.
- Visuals are being used: Renata, others
- Fewer late/missing submissions (on-ground team)

- Previous student:
  - “Now, when we talk to big language models like ChatGPT or DeepSeek, they can understand our mood, word choice, and situation, responding as though they’re one of our pals.”
- Previous student:
  - "Core difference: Naive Bayes asks which words are here. Claude asks what this sentence is actually saying. "

Feedback - Week 6

Assignment



Going well:

- Previous student: Good prompting technique

I am conducting an experiment on sentiment classification.
Please classify each of the following restaurant reviews as:
* Positive
* Negative
* Ambiguous
For each review:
1. Provide the sentiment label.
2. Explain your reasoning in 2–3 sentences.
3. Identify specific words or phrases that influenced your decision.
Here are the reviews:
1. “The food was absolutely delicious and the service was outstanding.”
2. “Best dining experience I’ve had all year. Highly recommend!”
3. “The food was cold and the waiter ignored us.”
4. “Terrible service and overpriced meals. Never coming back.”
5. “The food was okay, but the service could have been better.”
After classifying them, briefly explain:
The output should be in table view

Show less

15:32

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Assignment



Going well:

- Previous student, many others: attempt to compare in an organized fashion:

Table 1

Review Texts, Sources, Classifications, and Key Reasoning

#	Review (abridged)	Source	Classification	Key Reasoning
1	"Way better than you may have heard, this mesmerizing look at young Donald Trump and his legal dark prince Roy Cohn."	Metacritic (2024)	Positive	"Mesmerizing" and superlative framing ("way better") are strong positive anchors.
2	"The performances are	MUBI (2024)	Positive	Superlatives ("stellar,"

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Assignment



Going well:

- Renata: deeper exploration of the sentiment

Figure 1

Placement of the Five Reviews on a Negative-to-Positive Scale (Claude's Classification)

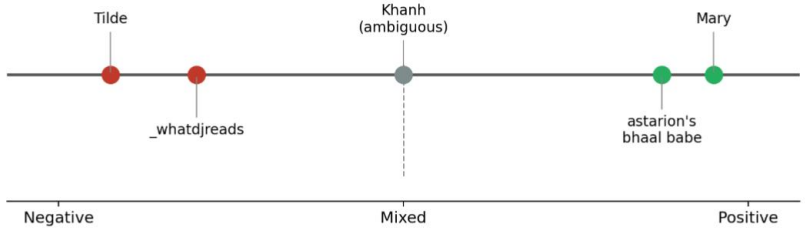
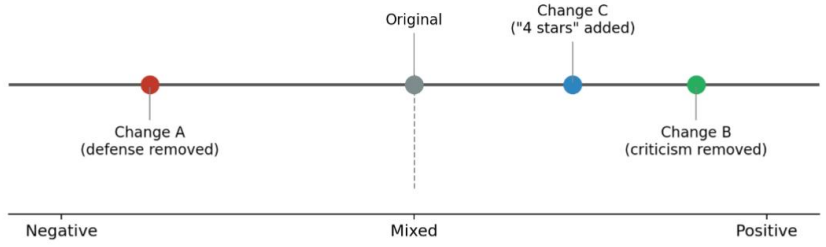


Figure 2

How Each Change Moved the Classification of the Ambiguous Review



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Assignment

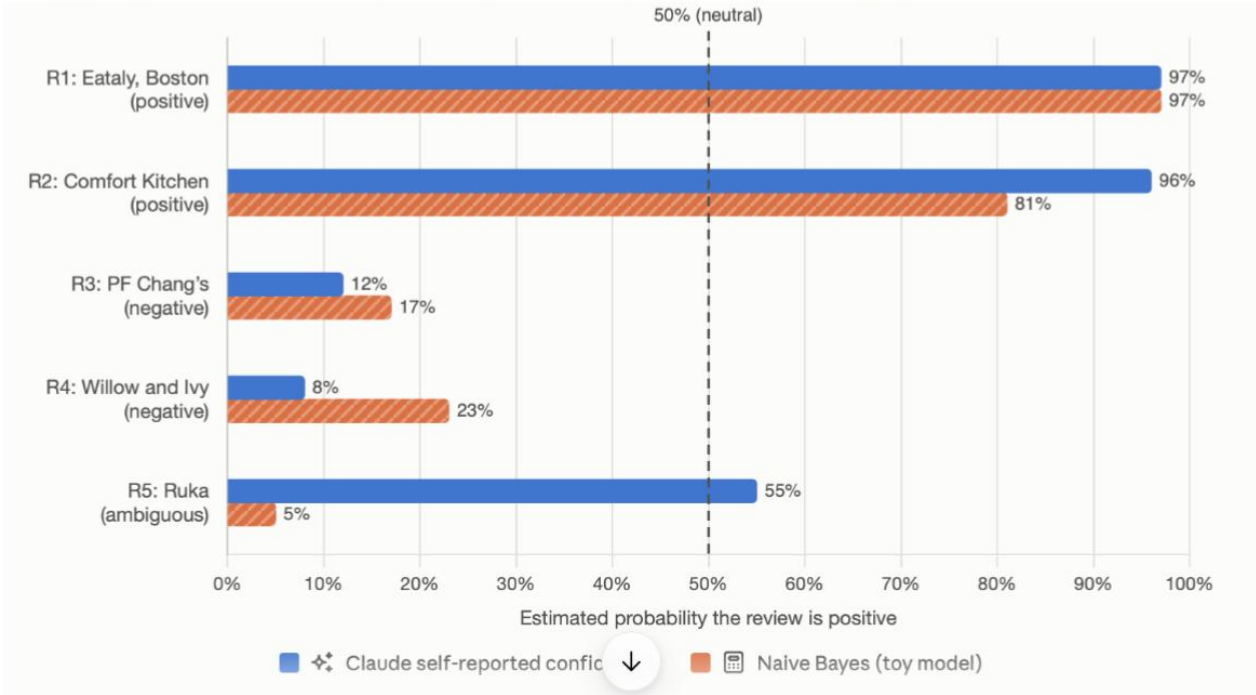


Going well:

- Mitzie: trained and compared a working Naïve Bayes model on the same reviews

Figure 1

Probability That Each Review Is Positive, Claude Versus Naïve Bayes



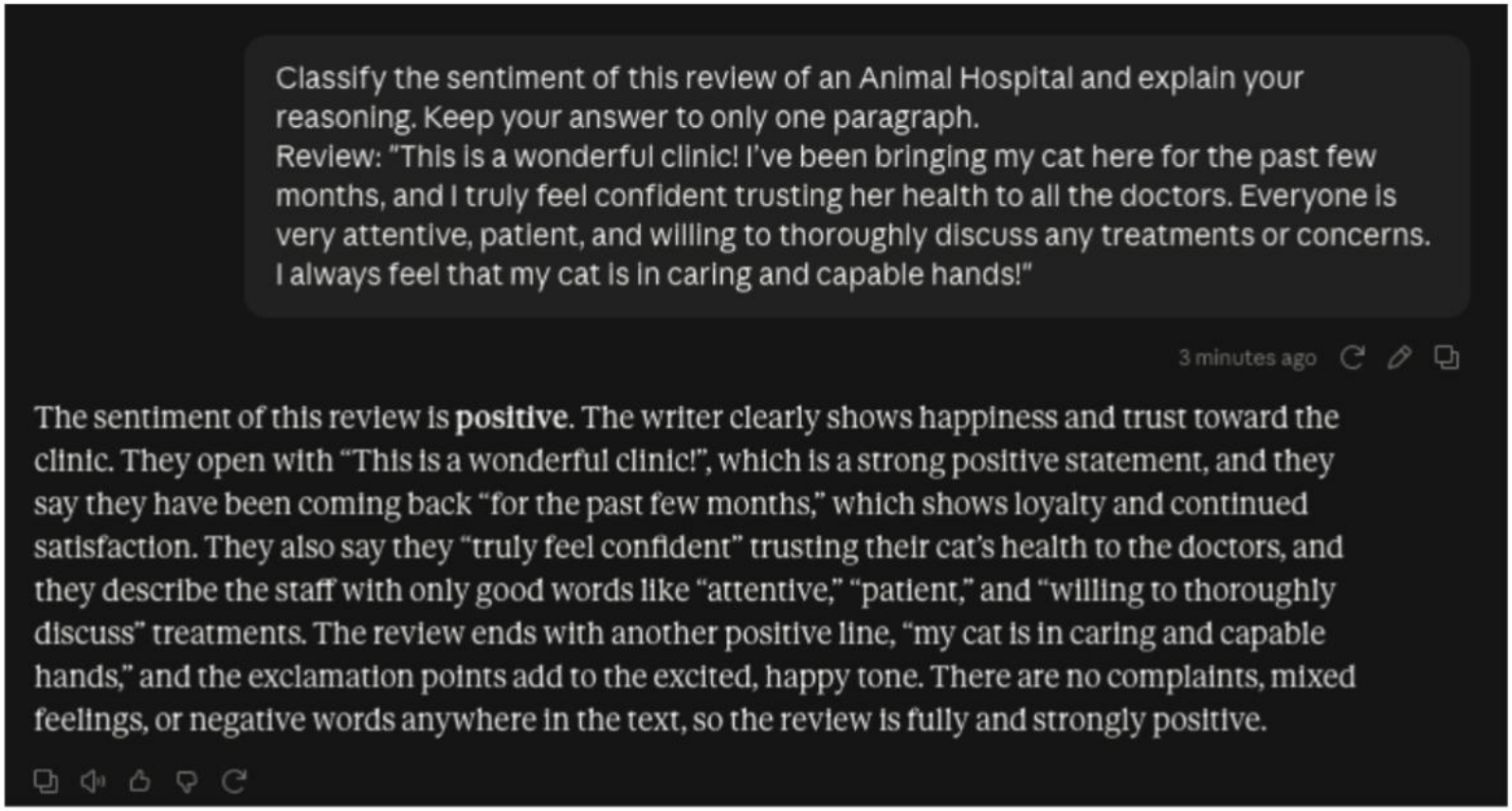
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Assignment



Going well:

- Luizza, others: showing the entire Claude interchange is helpful



Feedback - Week 6
Assignment (cont'd)

To improve :

- Strive for submissions to be self-sufficient. Why does it matter?
- A personal reflection on the quality of Claude's sentiment analysis would be good to provide
- Hallucinations, bias, etc., with NLPs. How about humans? – Autism, dementia, speech/mental disorders, forgetfulness. How about biases in humans?

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Assignment (cont'd)

My experiments with Claude and your reviews (last semester):

How about "The food was fine and the portions were big, but the wait was long
and the place was loud." for both the default and Naive Bayes methods.

Default Sentiment Assessment

Sentiment: Mixed/Slightly Positive

Score: 0.15

Assess the sentiment of the review "Amazing ramen—rich broth, perfectly chewy noodles, and the staff were genuinely kind. I'm coming back."

Sentiment: Strongly Positive

The review expresses enthusiasm across three dimensions: food quality ("amazing ramen," "rich broth," "perfectly chewy noodles"), service ("staff were genuinely kind"), and intent to return ("I'm coming back"). The language is warm, specific in its praise, and signals high satisfaction with no negative notes whatsoever.

Provide a score from -1 (strongly negative) to 1 (strongly positive)

Sentiment score: 0.95

The review is overwhelmingly positive with enthusiastic praise for the food and service, plus a clear statement of intent to return. It falls just short of 1.0 since the language, while very warm, isn't the most extreme level of praise possible (e.g., "best I've ever had," "life-changing").

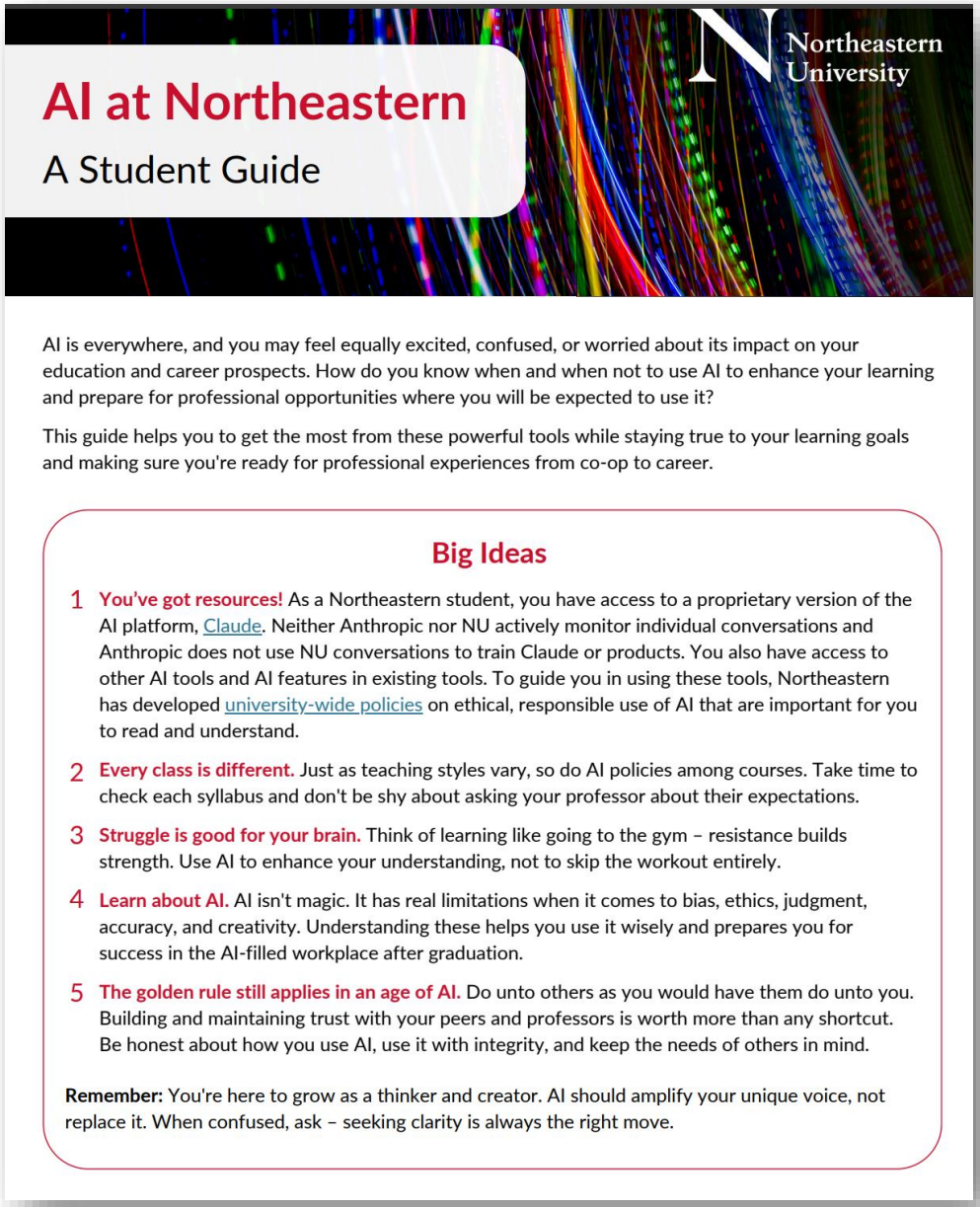
Assess the sentiment of the same review using the Naive Bayes method

Naive Bayes Sentiment Assessment

The Naive Bayes approach classifies sentiment by evaluating the probability of each word (or feature) belonging to a positive or negative class, based on learned prior probabilities from training data. Here's a walkthrough of how it would apply to this review:

AI at Northeastern

A Student Guide FYI:



AI at Northeastern

A Student Guide

Northeastern University

AI is everywhere, and you may feel equally excited, confused, or worried about its impact on your education and career prospects. How do you know when and when not to use AI to enhance your learning and prepare for professional opportunities where you will be expected to use it?

This guide helps you to get the most from these powerful tools while staying true to your learning goals and making sure you're ready for professional experiences from co-op to career.

Big Ideas

1. **You've got resources!** As a Northeastern student, you have access to a proprietary version of the AI platform, Claude. Neither Anthropic nor NU actively monitor individual conversations and Anthropic does not use NU conversations to train Claude or products. You also have access to other AI tools and AI features in existing tools. To guide you in using these tools, Northeastern has developed university-wide policies on ethical, responsible use of AI that are important for you to read and understand.
2. **Every class is different.** Just as teaching styles vary, so do AI policies among courses. Take time to check each syllabus and don't be shy about asking your professor about their expectations.
3. **Struggle is good for your brain.** Think of learning like going to the gym – resistance builds strength. Use AI to enhance your understanding, not to skip the workout entirely.
4. **Learn about AI.** AI isn't magic. It has real limitations when it comes to bias, ethics, judgment, accuracy, and creativity. Understanding these helps you use it wisely and prepares you for success in the AI-filled workplace after graduation.
5. **The golden rule still applies in an age of AI.** Do unto others as you would have them do unto you. Building and maintaining trust with your peers and professors is worth more than any shortcut. Be honest about how you use AI, use it with integrity, and keep the needs of others in mind.

Remember: You're here to grow as a thinker and creator. AI should amplify your unique voice, not replace it. When confused, ask – seeking clarity is always the right move.

Final Team Project

Reminder not to delay the work.

Topics Covered

Course Tracker

Week #	Topic
1	Overview of AI, history, definitions, and key concepts
2	Knowledge Representation and Reasoning
3	Introduction to Machine Learning
4	Neural Networks & Deep Learning
5	Distributed AI & AI Platforms
6	Natural Language Processing (NLP)
7	**Computer Vision**
8	Robotics & Embodied AI
9	AI & Creativity
10	Applications of AI – Healthcare, Finance, Education, Retail, Marketing, Entertainment
11	Applications of AI – Biomedical, Agriculture, Transportation, Mining, Manufacturing, Energy
12	AI and Society, Bias & Fairness in AI
13	The Future of AI
14	No Lecture - Final Project Report & Presentation are Due